Popeye: Hand Pose and Contact Force Estimation from Forearm Surface Imaging

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***Abstract*—Manipulation data from human hands is a valuable source of supervision for robot policies, but the methods for capturing it trade measurement fidelity against interference with the hand. Gloves and exoskeletons record joint angles and contact force directly but are bulky and push demonstrators towards slow, unnatural motions. Meanwhile, external camera-based approaches leave the hand bare but fail under occlusions that are common in natural hand use and typically recover no contact information. Instead, we leverage musculoskeletal anatomy to estimate hand pose and contact force from the forearm. We present *Popeye*, a forearm-worn device that estimates hand and wrist joint angles as well as per-finger contact force, with nothing worn on the hand and without requiring the hand to be visible from an external camera. The extrinsic muscle bellies that drive fingers and wrist are located in the forearm, and their contractions deform the forearm skin. We image this skin surface with four RGB cameras through a textured elastic bandage, alongside readings from an on-board IMU. Our device predicts finger joint angles to a mean geodesic error of 9.27° and wrist orientation to 13.50°, detects fingertip contact at an average precision of 0.823, and tracks force magnitude at a correlation of $r = 0.725$. The estimation pipeline is deliberately lightweight, using simple image feature extractors, a linear dimensionality reduction, and a compact multilayer perceptron, so predictions can run in real time rather than as an offline post-processing step. Popeye is built from off-the-shelf and 3D-printed components costing $< \$200$, and its design, code, and data will be made available open-source.**

I. Introduction

Human hands perform dexterous, contact-rich in-hand manipulation effortlessly, and this routine behavior can be a valuable source of supervision data for learning robot manipulation policies. However, natural in-hand human manipulation data is hard to acquire at scale. Simulation offers large volumes of data at low cost, but first requires careful tuning of contact, compliance, friction, and hardware parameters to transfer well in the real world [1, 2]. Human manipulation recorded in egocentric or internet video datasets captures natural in-hand dexterity in abundant, diverse environments, but often only coarse arm and hand motion can be extracted from these videos [3, 4]. Teleoperated leader-follower systems can produce more robot-ready manipulation data but require dedicated setups that can be hard to scale for high degree-of-freedom (DOF) end-effectors such as multi-fingered hands [5, 6].

More recently, sensorized gloves and hand exoskeletons have narrowed this gap by recording joint angles and contact forces directly on the hand as a user performs a task [7–11].

Authors are with the Massachusetts Institute of Technology (MIT). Website: https://patelvatsalv.github.io/popeye

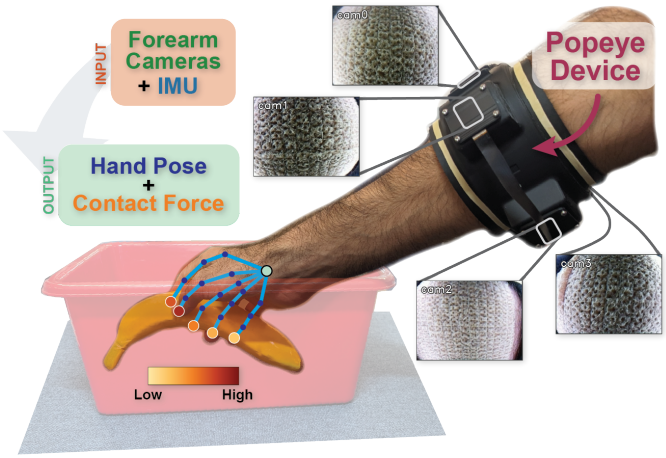


Fig. 1. The Popeye device worn on the forearm estimates hand and wrist pose as well as per-finger contact force from four inward-facing cameras that image the forearm surface through a textured elastic bandage, and from an on-board IMU. Without requiring visibility to external cameras or a bulky wearable on the hand, Popeye enables manipulation data collection with natural hand use, even under occlusion.

But these devices have to share the physical space with the hand, and their added mass and constrained kinematics push demonstrators towards slow, quasi-static motions rather than their usual fast, contact-aware behavior. Camera-based hand pose estimators avoid any interference with the hand entirely [12–14]. But these approaches require unobstructed, well-lit views of the hand, which natural in-hand manipulation motions regularly violate, as hands move inside boxes, under shelves, or behind grasped objects. Moreover, these vision-only approaches do not directly recover contact force information during manipulation.

So, existing methods of capturing human in-hand manipulation run into the tradeoff between measurement fidelity and interference with the hand: *approaches that directly measure the hand alter natural hand use, while approaches that leave the hand untouched fail during natural hand use.*

Instead, we propose observing the forearm rather than the hand itself. Much of the musculature that drives the fingers is not located in the hand. The extrinsic flexors and extensors, such as the flexors digitorum superficialis (FDS) and profundus (FDP), extensor digitorum communis (EDC), and flexor pollicis longus (FPL), have their muscle bellies in the forearm [15]. Muscles that move the wrist, flexor carpi radialis and ulnaris (FCR, FCU), and extensor carpi radialis and ulnaris (ECR, ECU), likewise have their bellies in the forearm. Finger and wrist motions, contact forces, and hand stiffness therefore

originate largely because of the length changes and activation of these extrinsic muscles. And since muscle fibers are nearly incompressible, contraction of a muscle increases its cross-sectional area, which can be observed as changes in shape of the forearm surface. We capture this surface motion by wrapping a textured elastic bandage directly over the muscle bellies to convert the bulging and contracting into observable pattern deformations, similar to vision-based tactile sensors that image a deformable membrane under contact [16].

We present *Popeye*, a forearm-worn device that estimates hand and wrist joint angles as well as per-finger contact force from images of the forearm surface. Four cameras view the bandaged forearm from four directions radially distributed around the forearm, and, along with an inertial measurement unit (IMU) sensor on the device, a compact model maps these inputs to hand state at every frame (Fig. 1). Imaging the forearm surface therefore offers two key capabilities over existing methods in one device: **(1)** hand and wrist pose as well as per-finger contact force are recoverable while the hand is obscured or out-of-sight, and **(2)** the hand remains bare during manipulation data collection and the operator can manipulate objects naturally.

The estimator model in this work is lightweight, consisting of an image feature extractor, a linear dimensionality reduction, and a compact multilayer perceptron. The hand state predictions can be run in real time rather than as an offline post-processing step common in transformer-scale hand reconstruction methods. Our forearm-worn device sits closer to the elbow than wrist- or hand-mounted devices, so it adds less inertia to the hand during manipulation. Across three participants, Popeye recovers finger joint angles to $9.27°$ and wrist orientation to $13.50°$ of mean error, detects fingertip contact at an average precision of 0.823, and tracks force magnitude at $r = 0.725$. The device is built from off-the-shelf and 3D-printed components costing $<$ \$200, and its design, code, and data will be released open-source.

A. Related Work

Vision-based hand tracking. Monocular hand reconstruction is accurate and simple to deploy [12–14], and multi-view variants have been used to label large manipulation datasets [17]. These methods require unobstructed views of the hand, are least accurate along the depth axis, recover no contact force, and at transformer scale, are typically run offline instead of real time.

Hand exoskeletons and gloves. Passive exoskeletons couple the human hand to an instrumented structure that records joint pose and contact [9, 11, 18], and tactile gloves place sensing elements closer to the fingers [7, 8, 10, 19, 20]. These approaches capture simultaneous pose and contact, and some provide haptic feedback, but they add bulk and often constrain hand kinematics. Most of these systems only instrument the finger joints, so wrist orientation and global position still require an external tracker.

Surface electromyography (sEMG). sEMG measures the electrical activation preceding muscle contraction, and has been used to predict continuous per-finger force from a wrist or forearm band paired with external hand pose detectors [21, 22], and to classify gestures and coarse hand poses [23]. Surface EMG can be demanding to deploy, as signals drift with muscle fatigue and electrodes are sensitive to placement, perspiration, hair, and electromagnetic interference [24].

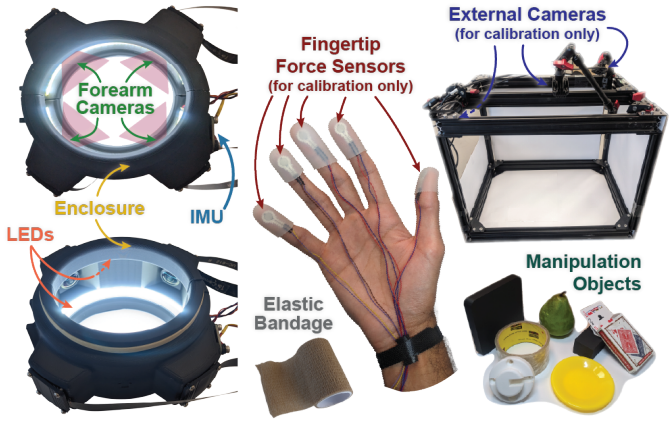


Fig. 2. The Popeye device (left), the calibration setup, and the objects used in data collection are shown. The two views of the device show the four camera locations, LED strips, and IMU mounted on the enclosure, which is worn over the elastic bandage on the forearm. The fingertip force sensors and external cameras are only used for ground-truth labels during calibration.

Wearable deep muscle imaging. Ultrasound and electrical impedance tomography (EIT) image the interior of the forearm and provide a rich biomechanical signal, including direct observation of muscle cross-section [25, 26]. Their drawbacks are hardware complexity and reliance on consistent acoustic or electrical contact with the skin, where even small changes in ultrasound probe position can lose features of interest [27].

Force and tactile myography. FMG/TMG measures skin deformation from contracting muscles underneath, typically with arrays of pressure-sensitive elements [27, 28], and compares favorably with sEMG [29]. Recent work uses several hundred pressure taxels at the wrist to predict continuous hand pose and contact force [30]. Optical variants use fiducial markers on the forearm [31], or a few photo detectors [32] for gesture classification. Our approach is similar in principle to these works, with our key insight being the use of cameras as a transducer for a high spatial density signal, which lets us recover continuous finger and wrist joint angles as well as per-finger contact force.

II. Device Design

The Popeye device is a rigid 3D-printed enclosure that wraps around the proximal forearm over a textured elastic bandage worn directly against the skin, and it carries four inward-facing cameras and an IMU (Fig. 2). The design constraints for our device were as follows: the extrinsic muscles are distributed radially around the circumference of the forearm, so multiple viewpoints are needed, and the device has to remain thin and light enough for long data sessions, which limits how far the cameras can stand off from the skin.

Forearm cameras. Four IMX219 camera modules are mounted radially at $90°$ intervals around the forearm, each

fitted with a 220° wide-angle lens and looking inward at the bandaged forearm surface from ≈ 15 mm. The cameras needed to stand off from the surface to be able to image the arc of the forearm that is not self-occluded by its own curvature, and we identified the standoff distance assuming an idealized circular forearm with a diameter of 80 mm such that the four cameras at 90° spacing could cover nearly the entire circumference. The four camera modules are routed through a Camarray HAT to a Raspberry Pi 4B, which streams them as a single stitched 1640×1232 frame (820×616 per camera) at 50 fps.

Textured elastic bandage. An elastic self-adhesive bandage is worn directly on the forearm skin under the enclosure. Bare skin is a poor imaging target as it can lack texture, and its appearance varies with hair, perspiration, pigmentation, and scarring. The bandage's knit pattern provides a high-contrast texture that is consistent across users. And since the elastic bandage is worn in tension, it follows the skin surface closely without restricting motion. Bulging and contracting of the muscles is thereby converted to in- and out-of-plane deformations that a camera can observe.

Device enclosure. The 3D-printed enclosure is held against the forearm with elastic bands wrapping over it and is designed so that the applied pressure on the forearm is not directly over the regions the cameras observe (as pressing on those regions would flatten some of the surface motion). The enclosure surrounds the imaged forearm surface, and two 5V white LED strips inside the enclosure provide constant illumination. This prevents changes in ambient light from affecting the feature extraction pipeline.

IMU sensor. A BNO055 nine-axis IMU is mounted on the outside of the enclosure and communicates with the Raspberry Pi over I2C at 100 Hz. The IMU readings provide the global orientation of the arm, which mainly helps the wrist orientation predictions.

Cost and fabrication. The device is built entirely from off-the-shelf electronics and 3D-printed parts, with a total bill of materials cost of $< \$200$. The design files and code will be made available open-source.

III. Hand State Estimation

A. Problem Formulation and Anatomical Basis

Given the four forearm camera images and the IMU orientation at time t, we estimate the hand and wrist joint angles q_t and the per-finger contact force f_t at that same instant. The model carries no state between frames and does not depend on its own previous outputs, so this is a per-frame estimation problem. Some image attributes such as flow features are differenced over a short backward window and detailed in §III-B below.

While a lot of hand behavior originates in the forearm muscle belly, these signals are not uniformly accessible. The *extrinsic* muscles are arranged in layers. The flexor digitorum superficialis (FDS), which flexes the proximal interphalangeal (PIP) joints, the wrist flexors and extensors, and the extensor digitorum communis (EDC) all lie superficially, closer to the skin. Whereas, the flexor digitorum profundus (FDP), which flexes the distal interphalangeal (DIP) joints, and the flexor pollicis longus (FPL), which flexes the thumb, sit deeper in the forearm [15]. Our device only images the surface, so the superficial group would be observed more directly. Independent contractions of the deeper muscles might not directly deform the skin, but these may still be weakly represented in our inputs because changes in the cross-sectional area of deeper muscles push on surrounding tissues to change the overall limb appearance [27].

Some hand motions have no forearm correlate at all. The *intrinsic* muscles, such as the dorsal and palmar interossei that drive finger abduction/adduction, and the thenar group that drives thumb opposition, sit entirely within the hand [15]. Any change in pose or force that these muscles generate in isolation produces no forearm deformation and is not recoverable by our device in theory. However, these motions do occur in conjunction with extrinsic muscle motions for a large variety of common grasping and manipulation primitives, so our model can still reproduce them by correlation but that is a prior learned from co-occurrence rather than a direct measurement.

So, the hand's 20+ DOFs are strongly coupled, both anatomically and in natural use. For instance, each digit has three flexion DOFs, from the knuckle outward: the metacarpophalangeal (MCP), PIP and DIP joints. Yet only two long flexors cross the digit, with FDS spanning MCP and PIP, and FDP spanning all three. The FDP tendons to the middle, ring and pinky fingers arise from a common muscle belly and are interconnected distally, which is why those digits cannot be flexed fully independently [15]. This anatomical coupling is also consistent with hand behavior. A small number of principal components account for most of the postural variance in grasping and everyday hand use [33, 34]. Thus, the mapping that our model has to learn is onto a low-dimensional manifold of reachable postures, and not onto an unconstrained 20+ DOF joint space.

Inferring hand state from forearm muscle activity is an established practice (§I-A), and our design addresses two of its main limitations. First, the modalities generally used are sparse transducer arrays of tens to a few hundred elements, which coarsely sample the skin surface [30]. Second, forearm skin deformation is ambiguous between a joint that moved and a muscle that contracted against an unknown external load, and this ambiguity is a known source of error under contact loads [27]. We address the first with cameras, which sample the surface far more densely than a sensor array of comparable complexity and resolve in- and out-of-plane motions, and the second by estimating pose and contact force jointly from a shared representation, so the loading condition is part of what the model predicts and not an unmodeled error. Our approach of using fingertip force sensors for ground-truth data does not measure co-contraction of antagonist muscles that stiffen a joint without changing its angle or net contact force, so hand impedance does not appear in our output but controlled perturbation studies could be run to capture these since the forearm does carry this signal [35].

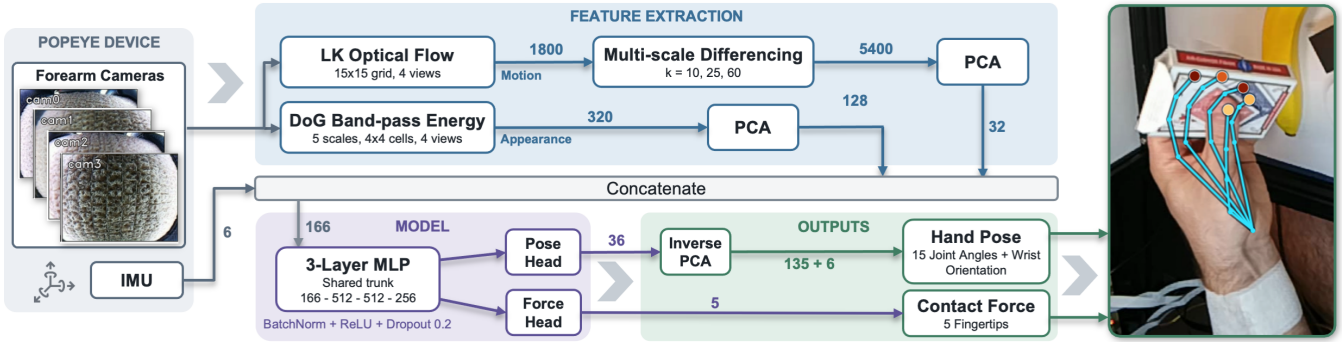


Fig. 3. The deployed estimation pipeline, where forearm cameras and IMU are the only inputs. Motion and appearance descriptors are extracted from the forearm images, reduced with PCA, and concatenated with the IMU orientation. The concatenated input is mapped to hand joint angles and fingertip force by a small MLP model with separate heads.

B. Estimation Pipeline

The deployed pipeline is summarized in Fig. 3. It is deliberately lightweight and consists of two image descriptors, a linear dimensionality reduction, and a small multilayer perceptron (MLP) with separate pose and force heads.

Image features. We extract two complementary descriptors from every stitched forearm frame. The first is a motion descriptor: cumulative Lucas-Kanade (LK) optical flow at a 15×15 grid of points in each of the four views, each point carrying a two-component displacement, for $15 \times 15 \times 2 \times 4 = 1800$ values per frame. Cumulative flow drifts over time, so the model takes a temporal difference at three scales ($k = 10$, 25, 60 frames at 50 fps). The second is an appearance descriptor: difference-of-Gaussians (DoG) band-pass energy, a common texture representation in computer vision [36, 37], which measures the bandage pattern's texture directly. Each view is blurred at $\sigma = 2.5$, 5, 10, 20 and 40 px, and consecutive blurs are differenced into four band-pass images plus a coarse residual. Each is then squared and averaged over a 4×4 grid per view, and passed through $\log(1+x)$, for $5 \times 16 \times 4 = 320$ values per frame. The two descriptors are complementary in theory, since a differenced motion descriptor drops to zero when the forearm surface is stationary even though the hand is still holding a posture, while the appearance descriptor is absolute and reports that posture directly.

Forearm orientation. The IMU quaternion is converted to the continuous 6D rotation representation given by the first 2 columns of the corresponding rotation matrix, contributing 6 values per frame.

Dimensionality reduction. We fit principal component analysis (PCA) separately on the motion and appearance descriptors of the calibration data, reducing to 32 and 128 components respectively. The components are not whitened, so each block retains the scale of its own eigenvalues. Concatenating the 2 reduced blocks with the 6 orientation values gives a 166-dimensional input vector.

Model. The MLP is a shared trunk of 3 fully connected layers ($166 \to 512 \to 512 \to 256$, each with batch normalization, ReLU and dropout of 0.2) feeding two heads. The pose head is a single linear layer producing 36 values, and the force head has one hidden layer ($256 \to 256 \to 5$, with ReLU and dropout), giving one output per fingertip. The trunk is already optimized against both pose and force losses in the same backward pass, so pose-relevant features are already available to the force head.

Targets. The pose target is the MANO hand pose [38], represented as rotation matrices for the 15 finger joints, flattened to 135 values and reduced by PCA to 30 components. The wrist orientation is an additional 6 values in the same rotation representation. The force target is the conductance of each of the 5 fingertip sensors, zeroed by the 5th percentile of each sensor's own conductance during a one-minute segment. The force head produces a single continuous value per finger, and the contact detection threshold is defined as a rise of 0.01 mS above zeroed baseline. Pose targets come from the multi-camera rig and force from fingertip sensors (§IV-A).

Training. We train with an L_1 objective applied blockwise to pose and force outputs, with the force block weighted $4\times$ relative to pose (Adam, learning rate 10^{-3}, batch size 64, weight decay 10^{-5}, at most 50 epochs). The reported metrics are on test segments, which did not influence the weights or the epoch selection. Finally, the predicted pose components are mapped back through the inverse PCA to recover joint rotations. No temporal filtering is applied at any stage and ground-truth pose and force are also left unsmoothed.

Per-user calibration. Both the PCA bases and the network are fitted per user. Calibration data is recorded in one sitting of 25 minutes while the participant manipulates objects freely without any scripted gesture sequences. This per-user fit is a common design assumption for such a deployed device that calibrates to each wearer.

Runtime. Feature extraction and inference take ≈ 30 ms per frame (33 fps) on a laptop CPU, of which the network itself accounts for 0.16 ms and the multi-scale appearance descriptor for ≈ 25 ms. Standard downsampling of that descriptor can reduce it to ≈ 10 ms. So our system can run online on the captured stream rather than as an offline post-processing step, which matters for teleoperation or prosthetic control.

IV. Experimental Evaluation

A. Experimental Setup

We obtain ground-truth hand pose from three OAK-1 Wide cameras viewing the hand from different angles (Fig. 2) at 1280×720 and 30 fps. In each view, the hand bounding

TABLE I
HAND POSE AND CONTACT FORCE PREDICTION ACCURACY

	Finger Error (deg) ↓		**Wrist Error** (deg) ↓		**Contact AP** ↑		**Force** r ↑
Participant	**Ours**	Constant Pose	**Ours**	Constant Pose	**Ours**	Chance	**Ours**
P1	**8.52**	11.55	**9.97**	28.51	**0.908**	0.335	**0.778**
P2	**10.05**	14.32	**17.17**	42.63	**0.742**	0.284	**0.616**
P3	**9.25**	13.18	**13.36**	47.00	**0.818**	0.264	**0.780**
Mean	**9.27**	13.02	**13.50**	39.38	**0.823**	0.294	**0.725**

box is detected with MediaPipe [14] and the MANO model is fit on the cropped image with HaMeR [12]. The three views are fused into one ground-truth hand pose per frame with confidence-weighted averaging, where the confidence combines palm frontality and 2D re-projection error (each view's fitted 3D joints projected back on the same view and compared against MediaPipe's 2D points). The forearm and external OAK camera streams are aligned at the nearest host timestamp. Not all three views yield a usable hand pose in every frame (60.5% of frames on average), and frames with no label are dropped from fusion, training, and evaluation.

The three-camera rig only provides the ground-truth pose supervision and evaluation metrics, and is not required when the Popeye device is deployed. We use this vision-based calibration pipeline for its accessibility and simple hardware requirements, but our reported pose errors are also bounded by the accuracy of this labeling pipeline.

Ground-truth contact force comes from 5 force sensitive resistors (FSRs) under silicone fingertip caps. The FSRs are routed through an ADS1115 ADC over I2C to the Raspberry Pi. Thin 30 awg cables are used to connect to the FSRs and routed over the palm and through a wrist strap so the fingers stay free to move. We report force as conductance instead of newtons, because each FSR has its own conductance-to-force response that can change with use and depends on how load is distributed on the fingertip. These sensor cables and the fingertip covers do marginally degrade HaMeR's detection rate (by roughly 7-10 percent points at worst), but the sensors are required to produce fingertip force labels. This trade-off to obtain force labels is inherent to the calibration rig and not our device, and further reinforces the motivation to leave hands bare during data collection.

We recorded three participants, each in a single, continuous 30-35 minute don session on their dominant hand (P3 left-handed) and recorded in one-minute segments. The first 25 minutes of data are used to train a model and the remaining 5-10 minutes to report the evaluation metrics, so the segments in this test set are strictly later in the recording than any training segments. Random splits of frames into train and test sets drop mean finger error by $\approx 3^\circ$ likely due to temporal leakage and frame adjacency, so in this work, we only report metrics on frames collected *after* training. Participants manipulate a variety of objects (Fig. 2) freely with no scripted sequences of poses or gestures. They are instructed to ensure that contacts on the hand involve the fingertips so that the FSRs can capture the force label.

Finger pose error is reported as the geodesic distance between predicted and ground-truth joint rotation matrices, averaged over the 15 finger joints in the MANO model, and wrist pose error is the same distance computed separately. We use geodesic instead of flexion-only error because it does not specify a joint axis, which better represents joints that are not exact hinges. For comparison, across our participant data, geodesic error is within 0.25° of flexion-only root-mean-square error (RMSE) but 2.3° above flexion-only mean absolute error (MAE). The reference baseline is a constant hand pose, determined as the mean pose over the training set. Contact detection is defined as a rise in conductance of 0.01 mS above zero (although reported metrics are stable across a $10\times$ range of this threshold) and is scored using average precision (AP), the area under the precision–recall curve. The contact baseline reported is prevalence of contact, which is the AP metric evaluated for predicting contact entirely at chance. Force magnitude is scored by Pearson correlation r between measured and predicted fingertip conductance, computed per minute of recording and averaged over the five fingertips. Repeating training across three random seeds on all participants gives a spread of 0.15° in finger error, 0.71° in wrist error, 0.014 in AP and 0.018 in r, so changes smaller than these are not resolvable.

B. Results

Table I reports the evaluation metrics for all 3 participants, on the test segments recorded after all training data. Finger pose averages 9.27° of error and wrist 13.50°. Contact is successfully detected at $\mathrm{AP} = 0.823$ and the predicted force magnitude tracks the FSR's measured output at $r = 0.725$. Fig. 4 shows estimated pose and force across different participants, objects and grasp types, and Fig. 5 shows the predictions tracking the finger and wrist trajectories and the contact force over a test segment. For comparison, recent ultrasound approaches can be more accurate on hand pose-only tracking, at substantially higher hardware cost and complexity [25], although the devices reviewed in that work span 7.35 to 22.58° RMSE for tracking hand DOFs on limited gestures. Contact detection from sEMG is in a similar range to ours [22], on different tasks and contact definitions.

The differences in metrics between participants seem to trace to the data collection rather than the model alone. P1 had the lowest pose error and also the lowest baseline error. P3 only had 51.0% of frames with all three views detecting the hand (versus 68.5% for P1), and on these frames, only trails P1 in finger error by 0.16°. P2 and P3's wrist error is also attributable partly to their larger range of motion in their respective sessions, as seen in the larger constant pose error.

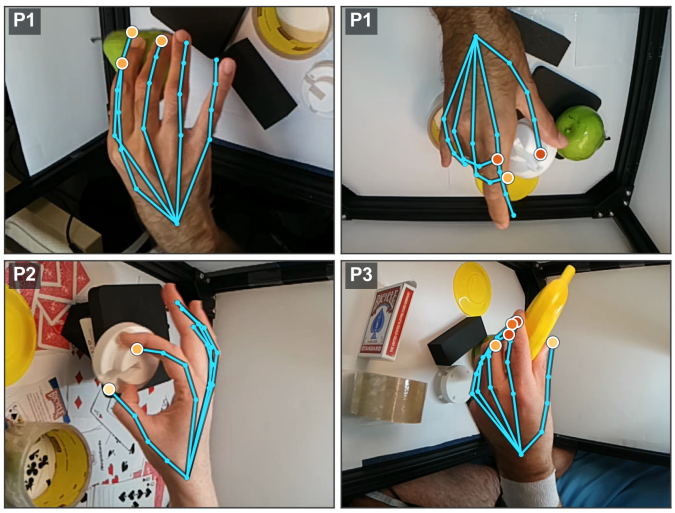


Fig. 4. Predicted hand pose and fingertip contact across the 3 participants. The cyan skeleton overlay is the predicted finger and wrist pose, and colored dots denote predicted contact force on the fingertip (darker for higher force). These snapshots are from separate recordings after calibration with the ground-truth force sensors removed so they do not contribute to reported metrics.

Prediction error also seems to fall with the number of views that detect the hand. On frames where all three views detect the hand, the finger error per participant is $8.08°$, $9.66°$ and $8.24°$, although this could also be because these frames correspond to easier hand poses where the hand is fully visible, so this only reflects an upper bound on the contribution of ground-truth pose label noise. Wrist orientation is the hardest output to predict throughout, which is also consistent with out-of-plane rotations being one of the least constrained components of monocular hand reconstruction [12]. However, wrist error from our model falls furthest below its constant pose baseline of any output, and ablations show this comes mostly from the cameras instead of the IMU (§IV-C).

Table II lays out the metrics per finger. Contact prevalence varies widely across the hand (from 8.4% for pinky to 51.6% for thumb), and as such, the detection AP metric should be read against the prevalence in the data collected. And assessing the raw AP against chance ranks the fingers similarly as the continuous force correlation does. However, the pinky's high r is pulled up by P1's 0.96, computed at only 2% contact prevalence.

Per-finger pose error is highest on the thumb and the pinky, but those two digits also have the highest constant-pose baselines, so these digits seem to move the most in the data collected. The joint-level pattern is even more informative. The MCP error averaged over participants is $10.69°$ (baseline $14.63°$) whereas PIP error is $8.86°$ (baseline $12.74°$). So the PIP joint is better predicted in every participant, which fits the anatomy, since PIP flexion is driven primarily by muscles in the forearm while the MCP joint takes some contributions from the interossei and lumbricals inside the hand. The margin between extrinsic- and intrinsic-driven DOFs was still small though, which suggests that our model is learning on the low-dimensional manifold of reachable postures described in §III-A, rather than resolving each joint independently.

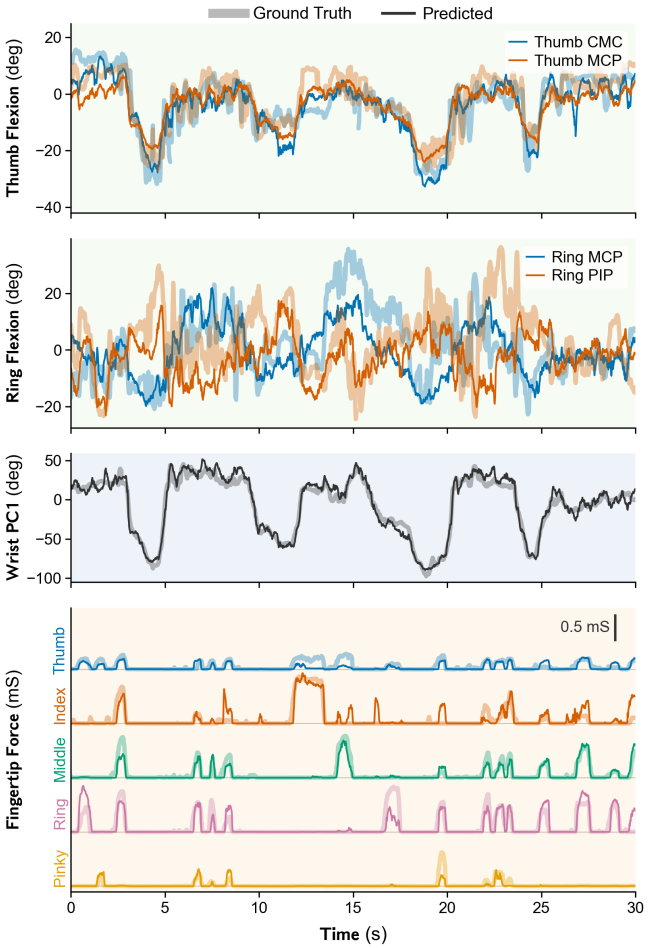


Fig. 5. Predicted and ground-truth hand state of a test segment on P3. The finger joint rotation traces are shown in their primary flexion axis, and the wrist about the principal axis of its rotation in this segment. Fingertip force values are shown in conductance (mS) units of the FSRs.

Pose error rises during contact events by 0.36 to $0.98°$ for the fingers and 0.49 to $1.34°$ for the wrist. We cannot separate where the model degrades versus where the ground-truth pose prediction degrades directly. The HaMeR hand pose labels are least reliable in the frames where the hand is occluded by an object or has overlapping fingers. The force labels are unaffected because the FSRs measure contact directly.

Accuracy does not decay substantially over the duration of data collection. Training each participant's model only on their first 20 minutes of data and evaluating on later segments (from 0.5 to 10 minutes after the training data ends), we find the finger error slopes for each participant of $+0.22$, -0.82 and $+1.04°$ per ten minutes. The inconsistent sign indicates some robustness to drift and device slippage. Wrist error slopes ($+1.40$, $+0.07$ and $+4.18°$ per ten minutes) do trend slightly upward, which may be from drift in the IMU's angle estimates.

Lastly, we evaluated the calibration curve by measuring prediction accuracy on the same test set with increasing amounts of training data. Ten minutes of a participant's data reaches $11.89°$ of finger error and a force correlation of $r = 0.484$, fifteen minutes $10.51°$ and $r = 0.623$, and the full twenty-five minutes reaches $9.27°$ and $r = 0.725$ (Table

TABLE II
EVALUATION METRICS PER FINGER AVERAGED OVER PARTICIPANTS

Finger	**Pose error** (deg) ↓		**Contact AP** ↑		**Force** r ↑
	Ours	Const. Pose	**Ours**	Chance	**Ours**
Thumb	10.43	13.88	**0.901**	0.516	0.643
Index	8.62	12.16	0.723	0.238	0.690
Middle	8.79	11.92	0.866	0.388	0.707
Ring	**8.31**	12.04	0.839	0.244	0.763
Pinky	10.21	14.24	0.785	0.084	**0.820**

TABLE III
INPUT ABLATIONS AVERAGED OVER PARTICIPANTS

Ablated	**Dim**	**Finger** (deg) ↓	**Wrist** (deg) ↓	**Contact AP** ↑	**Force** r ↑
−Appearance	38	**+1.69**	**+5.10**	**−0.193**	**−0.183**
−Motion	134	+0.35	+1.62	−0.020	−0.011
−IMU	160	+0.33	+2.82	−0.010	−0.009

I). Contact detection calibrates much faster than pose, and reaches $\mathrm{AP} = 0.634$ with just five minutes of data. When calibration data is scarce, the Popeye device could serve as a contact detector alongside external camera-based pose estimation. None of the reported metrics had flattened by 25 minutes, so more calibration data would still help. One source of more data would be a pooled training set across participants, although a model trained on one participant did not immediately transfer to another, and aligning data across people and forearm positions will be explored in future work.

C. Input Ablations

Table III reports metrics for ablations on the three input blocks. Each ablated version is a retrained model per participant. The appearance descriptor is the primary input by a wide margin and removing it costs $+1.69°$ of finger error and 0.193 of contact AP. The IMU primarily functions to support wrist angle predictions ($+2.82°$ of wrist error without IMU) and has marginal effect on finger and force predictions.

We had expected the two image descriptors to be complementary, with the motion descriptor carrying movement and the appearance descriptor carrying static information. The appearance descriptor seems to mostly subsume what optical flow has to offer, but we keep flow in our model since it is cheap to compute and still provides a small improvement. Future work will identify feature extractors that cover the appearance descriptor's failure modes and characterize different representations.

V. DISCUSSION AND FUTURE WORK

We presented Popeye, a forearm-worn device that recovers hand and wrist pose along with per-finger contact force from images of the forearm surface and an IMU, with nothing worn on the hand and no requirement that it be visible. Across 3 participants, the device predicts finger pose to a geodesic error of $9.27°$, wrist pose to $13.50°$, detects contact at $\mathrm{AP} = 0.823$, and tracks force magnitude at $r = 0.725$. It can be less precise than gloves or exoskeletons that instrument the hand directly, but it leaves the hand bare for natural manipulation and is built from off-the-shelf parts, making it a more practical way to collect contact-rich manipulation data at scale.

A. Limitations

Intrinsic muscle-driven hand DOFs are recovered by our model only by correlation as they produce no independent forearm surface change (§III-A), so we would expect them to degrade for hand configurations outside the calibration data. Calibration costs roughly 25 minutes of recording and must be repeated after each don cycle, or any change in position on the forearm from a large perturbation or adjustment, because the mapping from surface appearance to hand state is specific to where the cameras sit relative to the underlying muscles. For reference, comparable wrist-worn systems collect several hours of data per user with motion-capture ground truth [30]. Similar to gloves and exoskeletons, our device also does not measure global hand position on its own and requires an external marker attached, although it does recover wrist orientation, unlike many gloves.

Our ground-truth pose is vision-based, so the reported errors are bounded by the hand estimation pipeline, which suffers under occlusions and when only a single view detects the hand. The calibration rig therefore cannot validate pose predictions during many contact-rich motions and more external cameras would help [17]. The force labels from FSRs can behave nonlinearly and only register force applied directly over the sensor. Finally, these results come from 3 participants in one donning each, and characterizing how performance varies across people requires more participants and don cycles.

B. Future Work

We hope to collect a large manipulation dataset with this device and train robot policies on the resulting force-aware demonstration data. Pooling data across participants would improve accuracy, and a model pre-trained across users would shorten per-user calibration and generalize across don cycles. We plan to locate the device more consistently on the forearm using the IMU readings, landmarking features on custom patterned bandages, and palpable anatomical landmarks such as the medial epicondyle on the elbow.

On the sensing side, better camera coverage of the forearm and complementary modalities that measure deeper muscle activity would help. Surface EMG would add muscle activations on top of the muscle geometry we already measure. Because our device leaves the hand bare, its predictions could also be fused with pose from external cameras when the hand is visible. Better ground truth would raise the ceiling for performance too, and we will explore motion capture for pose and distributed contact sensing over the whole hand.

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